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Automated knowledge extraction from marine accident reports using large language models: Graph construction and evaluation

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ABSTRACT

Marine accident investigation reports contain valuable yet underutilized information due to their unstructured nature. While knowledge graphs (KGs) offer a promising solution for structured modeling and quantitative reasoning based on these reports, the construction of high-quality, domain-specific KGs remains technically challenging. To enable more efficient and scalable knowledge extraction in maritime safety domain, this study proposes an automated pipeline based on large language models (LLMs) which enhances existing approaches by customizing LLM prompting and evaluation frameworks. A chain-of-thought and one-shot prompting method is adopted to identify critical hazardous factors and their coupling relationships. Besides, a comprehensive quality assessment framework is developed to assess the accuracy and complexity of the generated KGs. Applied to real-world accident reports from maritime authorities, the proposed LLM-based method demonstrates substantial improvements in automation efficiency and extraction accuracy compared with traditional approaches. Furthermore, the quality evaluation framework provides complementary indicators to traditional metrics by reflecting the global semantic coherence and structural redundancy of the maritime safety KG. The findings offer actionable insights for deploying LLM-based tools to support data-driven maritime risk analysis enhancing transport safety management and decision making in complex operational environments.

1. Introduction

Maritime transport safety is influenced by a complex interplay of factors, including adverse weather conditions, vessel defects, human error, and operational mismanagement (Rawson et al., 2021; Wu et al., 2022). The occurrences of marine accidents are often characterized by uncertainty, and can result in severe consequences, affecting lives, assets, and the environment (Shu et al., 2025; Siqueira et al., 2022). Consequently, such accidents are typically sparsely and irregularly recorded over time and documented in unstructured formats with high-dimensional representations for post-incident analysis (i.e., accident investigation reports). While these reports contain rich information, their narrative style and domain-specific language pose significant challenges to effective data utilization (Yan et al., 2023).

To address these challenges, knowledge graphs (KGs) have emerged as a promising solution for modeling marine accident process by transforming unstructured text into structured semantic networks. They are able to support the identification of causal chains, control and mitigation of risk propagation, and prediction of the hazardous events (Liu et al., 2025). However, developing high-quality, domain-specific KGs remains a labor-intensive task. It demands significant domain expertise in both maritime operations and technical language interpretation, making it difficult to scale across large accident report collections.

Recent advances in generative large language models (LLMs) have opened new possibilities for automating this process. With the powerful capabilities of semantic understanding and logical reasoning, LLMs can perform key tasks, such as named entity recognition (NER) and relation extraction (Wang et al., 2023; Dagdelen et al., 2024). Yet, the application of LLMs in maritime safety contexts is still in its infancy. Moreover, their outputs are prone to pitfalls (e.g., hallucination, inconsistent coreference, and omission of critical context), potentially compromising the validity of the resulting KGs.

This paper addresses these challenges by investigating how LLMs can be effectively employed for automated knowledge extraction based on

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marine accident investigation reports from the United Kingdom (1990–2023) and China (2016–2024). This paper also investigates how to evaluate and optimize the quality of the generated KG. Ultimately, the proposed KG construction and evaluation framework is expected to enable scalable analysis of large volumes of marine accident reports and support systematic identification of risk factors to enhance safety in the maritime transport domain. The main contributions of this paper are:

(1) Proposing a scalable, LLM-based pipeline for constructing and evaluating KGs in the marine accident domain, reducing reliance on manual review.
(2) Developing a dual-perspective quality assessment framework, assessing both structural complexity and semantic accuracy, to systematically diagnose KG quality.
(3) Advancing the understanding of the mechanisms through which key configuration dimensions influence the KG quality, and introducing optimization strategies to mitigate LLM-related limitations in maritime safety applications.

The remainder of this paper is organized as follows: Section 2 reviews related work on KG construction and assessment methods in maritime safety domains and identifies key research gaps. Section 3 presents the proposed LLM-based KG construction and assessment framework. Section 4 demonstrates its application to real-world accident records from the U.K. and China. Section 5 further discusses the influence of configuration tuning and outlines an optimization strategy. Finally, Section 6 concludes the study and suggests future research directions.

2. Literature review

2.1. Marine accident analysis with knowledge graphs

Marine accidents are influenced by dynamic meteorological conditions (e.g., winds, waves and low visibility), vessel inherent defects, and operational or management failures (Luo et al., 2024; Yang et al., 2024; Ma et al., 2025). These incidents are often low in frequency but high in consequence (Chen, P. et al., 2022). Compared to other operational data, the accident records are sparse and predominantly unstructured, posing challenges for traditional accident analytical models such as Fault Tree Analysis (FTA) and Bayesian Network (BN) that demonstrate limitations in semantical representation and nuanced information storage (Li et al., 2024; Xiong et al., 2024). To address these issues, KGs have emerged as a promising approach for marine accident analysis.

KGs represent complex systems through networks of heterogeneous entities, attributes, and relations, making them particularly well-suited for modeling multi-factorial causes and cascading effects (Zarei et al., 2022). To accurately represent the marine accident process, customized concepts and entity types must be defined in addition to the general entity types (e.g., person, location, time, and organizations) used in open-domain tasks (Barron et al., 2024). For instance, Gan et al. (2023) developed an ontology of ship collision accidents integrating context-based and content-based metadata. Contextual elements, such as accident descriptions, responsibility, and administrative advice, were incorporated to preserve the original information of reports, while content elements including involved ships, location, causes and consequences, were identified to reflect multi-dimensional perspective of the accidents. Attributes of entities, such as the numerical hazard severity level and the enumerated event types, were used to provide structured variability across standardized entities (Zhu et al., 2024). As for relations, causal and spatial-temporal relationships are crucial in accident causation modeling. Xie et al. (2024) employed causal and sequential relations to link the vessel events and actions. Huang et al. (2025) constructed causal chains to capture the causal and temporal dynamics of event evolution in railway accidents.

By encoding the above elements into graphs, KGs transform unstructured narratives into machine-readable, semantically rich formats while preserving much of the original contextual details. Despite the promise, challenges remain in constructing high-quality domain-specific KGs due to the complexity of unstructured texts rich in technical terms (Liu et al., 2023). The time-intensive nature of knowledge extraction along with the high cost of expert annotation, continues to hinder their widespread application in maritime safety management.

2.2. Knowledge graph construction methods for the marine accident domain

A typical KG in the marine accident domain consists of two layers: a schema layer defining the conceptual structure and a data layer where case-specific information is instantiated (Zhang et al., 2019). Two primary ways of constructing KGs are employed: (1) top-down: an ontology is defined as the schema first and then used to guide knowledge extraction; (2) bottom-up: KG elements are extracted from data without schema constraints and then followed by schema abstraction. In domains with well-established domain knowledge, such as maritime safety, the top-down method enables the systematic formalization of existing expertise into reusable ontologies, thereby facilitating structured knowledge extraction and reasoning, while the bottom-up method is widely applied in open-domain KGs (Chen, T. et al., 2022).

Schema design in the maritime context is heavily dependent on domain expertise, requiring accurate definitions of entities, relations, and constraints unique to shipping operations and accident causation (Zhou et al., 2022). In contrast, the information extraction process, typically involving entity recognition and relation extraction, follows a more standardized pipeline. In the studies by Liu et al. (2024) and Huo et al. (2024), domain-specific KGs were constructed for the accident analysis in railway operations and construction sectors through a pipeline-based extraction approach. These works laid important foundations for structured knowledge representation in specific domains, although the extraction process still relied primarily on manual annotation and rule-based methods. With the growing volume of multi-source marine accident reports, there is an increasing need for automated, scalable, and robust information extraction techniques (Zhong et al., 2023). Deep learning and statistical learning models have exhibited satisfying performance in automated NER and relation extraction tasks. For instance, Liu and Cheng (2024) combined Bidirectional Encoder Representation from Transformer (BERT), Bidirectional Long Short-Term Memory (Bi-LSTM), conditional random fields, and a self-training strategy to develop a maritime-specific NER model that achieved satisfying performance. However, these enhanced pre-trained models often require annotated corpora and domain-specific retraining, limiting their adaptability and scalability. To alleviate the effort of annotation, unsupervised methods such as topic model have been applied in extracting structured knowledge from open-ended texts (Baburajan et al., 2022). Yet, such models may fall short in capturing complex event structures or semantic relationships.

Recent advancements in generative LLMs provide a promising alternative. By reframing KG construction as sequence-to-sequence structure prediction task, LLMs can better comprehend complex task-related knowledge with multi-head self-attention mechanisms (Ye et al., 2022). LLMs have been successfully prompted to carry out open-domain KG construction under Wikidata schema without annotated data (Feng et al., 2024; Abolhasani and Pan, 2024). Moreover, LLM-generated KGs serve as foundational components in advanced retrieval-augmented generation (RAG) techniques, such as GraphRAG (Edge et al., 2024) and knowledge-augmented generation (Liang et al., 2025). Beyond KG construction, LLMs have also demonstrated strong capabilities in bridging semantic understanding with physical reasoning in complex transportation systems (Liu et al., 2025). Given the complexity and domain-specific language of marine accident data, which includes technical terminology and regulatory references, manual KG construction and data annotation remain costly and inefficient. This

motivates the exploration of LLM-based KG construction for maritime safety analysis.

2.3. Quality assessment of domain knowledge graphs

Quality assessment of the developed KG is of significant necessity for ensuring its reliability and utility in downstream tasks. While manual evaluation remains the most accurate approach in theory, it is resource-intensive, non-scalable, and unsuitable for benchmarking across algorithms or large-scale datasets (Qi et al., 2022; Si et al., 2024).

One common strategy is the partial gold standard approach, where a subset of data is selected as test set and manually labeled to calculate standard metrics such as F1 score and area under curve (AUC) for knowledge extraction quality evaluation (Paulheim, 2016). However, this approach only assesses extraction accuracy and fails to capture broader dimensions of KG quality. To address this, scholars have extended general data quality measurement (i.e., intrinsic, contextual, representation, and accessibility) to KG quality evaluation (Xue and Zou, 2022). Besides, the structure and path characteristics of KG bring additional possibilities of developing dedicated dimensions and metrics. Metrics regarding abundant sub-division of ontology hierarchy, the number of attributes per class, class diversity, and overall graph complexity have been proposed in existing studies (Seo et al., 2022).

Despite various frameworks being proposed, many are either overly complex for practical use or lack scalability for large-scale KGs (Chen et al., 2019). Moreover, effective evaluation should be grounded in the domain-specific use cases. In marine accident analysis, particular emphasis should by placed on verifying the completeness and retrievability of triggering factors and risk propagation paths, which are central to safety diagnostics and causation modeling (Ma et al., 2022). Another concern is that the LLM-based knowledge extraction may introduce extra quality risks, such as hallucinated facts, coreference errors, and missing context. These risks highlight the need for assessment methods capable of detecting semantic inconsistence and verifying the fidelity of the extracted knowledge against the original text (Zhang et al., 2024).

2.4. Research gap and workflow

Despite growing interest in developing KGs for marine accident analysis, few automated methods have been tailored to handle the unstructured, technical, and narrative-heavy nature of accident reports. While recent advance in generative LLMs brings potential for schema-guided extraction, their application in the maritime safety domain remains underexplored. Based on the above review, three key research gaps are identified:

(1) The development of automated pipelines specifically tailored for structuring information from marine accident reports remains underexplored.
(2) Establishing a comprehensive and scalable quality assessment framework remains an important need for assessing the quality of domain-specific KGs and advancing their applications in the context of marine accident analysis.
(3) The impact of key configuration choices, such as decoding parameters, schema, and prompt design, on the quality of LLM-generated KGs has not been fully investigated.

To address these gaps, this study proposes an LLM-based framework for automated KG construction and quality assessment in marine accident analysis. The construction pipeline applies chain-of-thought and few-shot prompting strategies to guide the LLM to identify the entities and relations, and employs encoding model to generate embeddings for the extracted knowledge. In terms of evaluation, a dual-perspective framework is introduced. Complexity metrics assess whether the structural richness of the KG contributes meaningful insights for marine accident analysis. The accuracy assessment relies on LLM-aided text reconstruction and textual similarity scoring to evaluate the faithfulness of KG. By using these metrics, this study further explores the influencing factors and risks of LLM-based KG construction based on comparison analysis across various configuration settings on real-world marine accident corpora.

3. Methodology

3.1. Methodological framework

The proposed framework for constructing and evaluating a KG in the maritime safety management domain is illustrated in Fig. 1. It comprises four core modules, which are data collection, schema construction, KG generation, and KG quality assessment. This study primarily focuses on the application of LLMs to the KG construction and quality assessment processes.

First, marine accident data are sourced from official web-source databases provided and maintained by maritime authorities. The raw textual records are preprocessed and segmented into text chunks in a customized corpus for safety knowledge extraction. The chunks serve as inputs of the KG generation module.

Second, schema design is a critical prerequisite for KG development. It defines the scope of domain knowledge by specifying marine-and-safety related concepts and identifying potential entity and relation types. The schema provides structural guidance to the LLMs during knowledge extraction, enabling consistent and domain-aware outputs.

Third, the process of knowledge extraction is decomposed into a series of sub-tasks. Each sub-task is handled by a generative LLM (KG extractor) with task-specific prompt engineering. Based on the text chunks and schema, the model outputs subject-predicate-object (SPO) triplets, which are stored and visualized in a graph database. Moreover, their vector representations are generated by an embedding LLM (KG encoder) and stored in vector database to facilitate downstream tasks.

Fourth, a dual-perspective assessment framework for maritime safety management KG is proposed in terms of complexity and accuracy. Complexity evaluation focuses on identifying redundancy in entities, relations, and graph structure, ensuring the KG remains concise and free of trivial or unnecessary components. Redundancy refers to the presence of semantically equivalent entities, invalid or duplicated relations, and excessive structural elements that contribute little meaningful information to the knowledge graph (Zhong et al., 2023). To assess accuracy, a novel evaluation approach based on text reconstruction and semantic similarity analysis is introduced. A generative LLM (text reconstructor) is prompted to regenerate textual descriptions from sub-graphs corresponding to individual accident records. Semantic similarity between the original and reconstructed text chunks serves as a proxy for the fidelity of the KG. The assessment results guide iterative refinements to the schema and prompt templates, facilitating continuous improvement of KG quality in maritime safety management applications.

3.2. Data collection and chunk split

Marine accident investigation reports provide detailed descriptions of incident progression, contributing factors, and outcomes. These unstructured textual records often involve intricate relationships between entities and hazardous events, making them a valuable source for safety risk analysis. Marine authoritative bodies, such as Marine Accident Investigation Branch (MAIB) in the U.K.,[1] Maritime Safety Administration (MSA) in China,[2] and National Transportation Safety Board (NTSB)

[1] Source of MAIB reports: https://www.gov.uk/maib-reports.
[2] Source of MSA reports: https://www.msa.gov.cn/page/outter/shiguyufang.jsp.

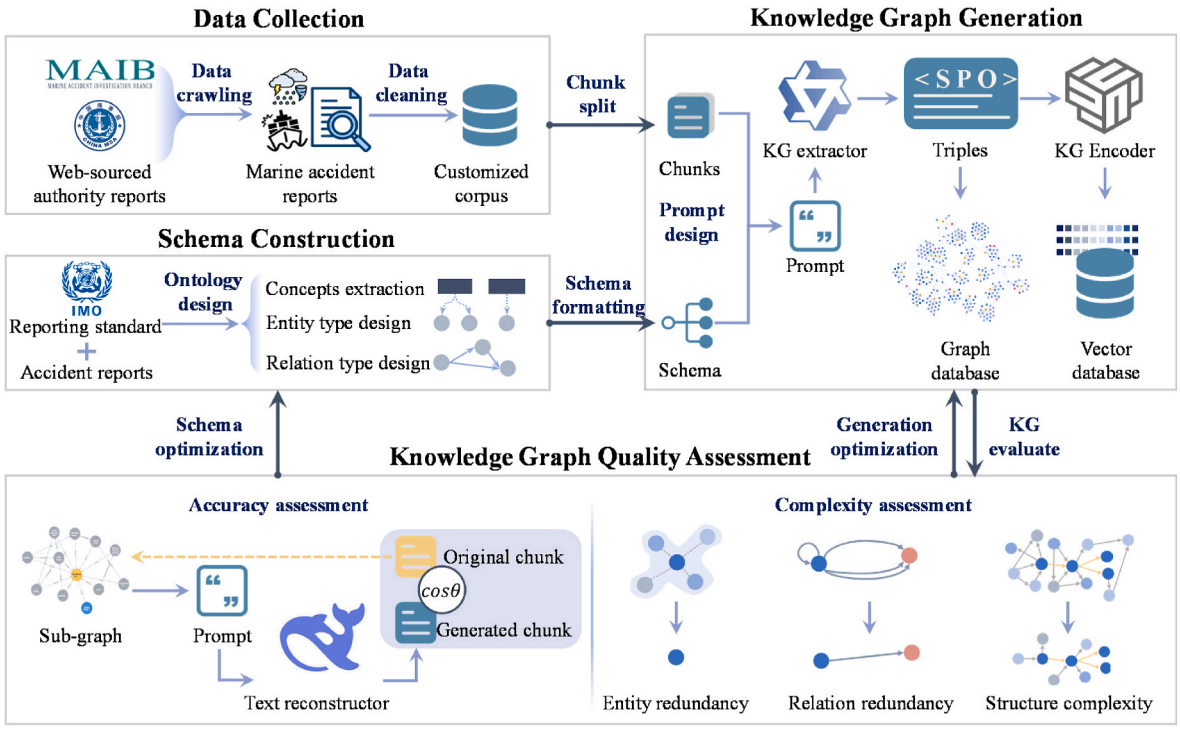


Fig. 1. Methodological framework of knowledge graph construction and evaluation in maritime safety management domain.

in the U.S.,[3] have been issuing the marine accident investigation reports, which serve as the primary data source in accident analysis.

In this study, a domain-specific corpus by focusing on weather-related marine accident reports is established and sourced from 172 MAIB reports (from 1990 to 2023) and 33 MSA reports (from 2016 to 2024). These two databases are selected due to their high-quality documentation and the ability to filter reports by keywords or accident category labels, which facilitates the identification of relevant reports. This choice of focusing on weather-related accidents is driven by both methodological and practical considerations. Methodologically, although the dataset coverage is limited, the designed schema still captures multiple causal dimensions including vessel characteristics, human factors, and operational contexts, which allows the framework to process diverse types of entities and relations in other maritime accident scenarios. Moreover, weather-related cases usually exhibit more uniform causal chains and more consistent contextual semantics, which facilitate precise schema design and enhance the LLM's extraction accuracy. Practically, extreme maritime weather has demonstrated significant impact on safety risk in recent years, underscoring the urgent need for action by the shipping and ocean communities (Brandt et al., 2024). Weather-related marine accidents are identified through abstract screening and keyword filtering. Relevant paragraphs describing accident summaries and causative analyses are collected to form the domain-specific corpus for KG development. A more detailed description of the MAIB and MSA datasets is provided in Section 4.

In the chunk split stage, each selected accident report is segmented into a single chunk representing the complete context of an individual accident. This design ensures that all extracted entities and relations within a chunk are semantically and contextually linked to the same case. By maintaining the integrity of accident-level context during knowledge extraction, the approach reduces the risk of cross-record semantic interference and enhances the accuracy and traceability of downstream KG construction and evaluation.

[3] Source of NTSB reports: https://www.ntsb.gov/investigations/AccidentReports/Pages/Reports.aspx.

3.3. Schema construction

The KG schema is designed to model the structural and causal logic of weather-related marine accidents, supporting downstream applications, such as risk pattern induction, responsibility identification, and resilience-oriented decision-making. The construction process follows a problem-driven and ontology-informed approach comprising the following five key steps.

(1) Objective Definition

The objective of the schema is to formalize key elements of weather-related marine accidents via the 5W1H framework (who, where, when, what, why/how) in a comprehensive and machine-interpretable form (Yang et al., 2011). The schema is hierarchically designed to incorporate the domain-relevant concepts, entity types, and relations. Table 1 presents the correspondence between schema elements and the 5W1H dimensions.

(2) Concept Layer Design

Top-level concepts (*Vessel*, *Time*, *Location*, *Weather*, *HazardousEvent*, and *Accident*) are identified through expert analysis of representative accident reports. These concepts serve as semantic anchors for defining subordinate entity types, thereby enhancing interpretability and structural clarity.

(3) Entity Type Specification

Subclasses of core concepts are organized as entity types based on frequently occurring elements in marine accident reports. For instance, *ShipName*, *ShipType*, *Cargo*, and *ShipOperation* are grounded under the concept of *Vessel*. For entity types with multiple but bounded categories, such as *ShipOperation*, *ShipResponse*, *ContributingFactor*, and *AccidentEvent*, enumerated attributes are introduced to standardize their ranges and reduce linguistic variability. These categories are informed by the International Maritime Organization's accident reporting guidelines

Table 1
Schema design for weather-related marine accidents knowledge extraction within 5W1H framework.

5W1H Component	Concepts	Entity types	Relations
Who: Identifies the vessels and other entities involved.	*Vessel*	*ShipName*, *ShipType*, *Cargo*, *ShipOperation*	<*ShipName*, *hasTypeOf*, *ShipType*> <*ShipName*, *carry*, *Cargo*> <*ShipName*, *engagedInOperation*, *ShipOperation*>
When: Indicates the time and date of occurrence.	*Time*	*Date*, *Timestamp*	<*ShipOperation*, *occurredOn*, *Date*> <*ShipOperation*, *occurredAt*, *Timestamp*> <*ShipResponse*, *occurredAt*, *Timestamp*> <*AccidentEvent*, *occurredAt*, *Timestamp*>
Where: Specifies the geographical location of the event.	*Location*	*Location*	<*ShipOperation*, *occurredAt*, *Location*>
What: Describes the nature of the marine accident.	*Accident*	*AccidentEvent*, *Casualty*, *CargoLoss*	<*AccidentEvent*, *causes*, *Casualty*> <*AccidentEvent*, *causes*, *CargoLoss*>
Why/How: Explains the causes, contributing factors, and sequence of events leading to the incident.	*Weather*, *HazardousEvent*	*WeatherType*, *WindForce*, *SeaState*, *Visibility*, *ContributingFactor*, *ShipResponse*	<*WeatherType*, *characterizedBy*, *WindForce*> <*WeatherType*, *characterizedBy*, *SeaState*> <*WeatherType*, *characterizedBy*, *Visibility*> <*WeatherType*, *causes*, *ShipResponse*> <*WeatherType*, *causes*, *AccidentEvent*> <*ShipName*, *exhibitedResponse*, *ShipResponse*> <*ContributingFactor*, *causes*, *AccidentEvent*> <*ShipResponse*, *causes*, *ShipResponse*> <*ShipResponse*, *causes*, *AccidentEvent*> <*ShipResponse*, *causes*, *CargoLoss*>

(IMO, 2014).

(4) Relation Specification

The relations are specified as SPO triplets as shown in the fourth column of Table 1. These relations are defined to capture both static and dynamic aspects of maritime operations and accidents. Static facts describe inherent properties of vessels (e.g., *hasTypeOf*, *carry*). Dynamic relations reflect the time-varying weather conditions and operational contexts (e.g., *characterizedBy*, *engagedInOperation*, *exhibitedResponse*). Spatial and temporal anchors (e.g., *occurredOn*, *occurredAt*) are introduced to provide contextual alignment with time and location of the dynamic events. The causal relations (e.g., *cause*) explicitly model the accident progression with causal chains, allowing the KG to represent the accident evolution paths among weather conditions, ship responses, contributing factors, and accident consequences.

(5) Schema Formatting

Given its role as a key input to LLMs, the schema should be both human-readable and parseable by machines. Thus, it is organized by JSON format in this study, which facilitates iterative schema refinement, programmatic manipulation and compatibility with LLM prompts. A preliminary schema for extracting weather-related marine accident knowledge is presented in Appendix A.

3.4. Knowledge graph generation

An automated pipeline for knowledge extraction and KG development is proposed with the aid of LLMs and alongside programmatic controls. Considering the complexity of knowledge extraction, it is decomposed into a series of sub-tasks: NER, entity standardization, and relation extraction, with each step handled by LLMs. This design improves the extraction reliability by simplifying the prompt in each step to avoid the distraction of LLM (Hwang et al., 2025). Besides, it enables stepwise quality control and optimization, in line with the principles of chain-of-thought.

3.4.1. Named entity recognition

NER aims to identify elements involved in each accident record. A one-shot prompt is designed to guide the generative LLM, referred to as the KG extractor, to extract entities and their attributes in accordance with the schema constraints. The prompt includes role and context instructions, the extraction schema, an annotated example, and output formatting requirements. An example of NER prompt is provided in Appendix B.

3.4.2. Entity standardization

Variability in the entity naming across documents may lead to redundancy of entities, negatively impacting the scale of KG, the calculation efficiency, and relation extraction accuracy. To address this issue, it is necessary to transform the entity mentions into normalized forms. The KG extractor is prompted to map overly specific terms to standardized equivalents (e.g., timestamps, measurement units, and common hazardous events). An example standardization prompt is included in Appendix B.

3.4.3. Relation extraction

The final step in the extraction pipeline involves identifying the SPO triplets among extracted entities, where the relation types are strictly in consistent with the scheme. Emphasis is placed on producing well-structured JSON output, with instructions to validate and, if needed, revise the result. Appendix B includes a sample prompt for relation extraction.

3.4.4. LLM decoding parameter configuration

In addition to prompting, the LLM performance is also influenced by decoding parameters, which should be specified to control the randomness and diversity of the outputs when calling LLM. Specifically, this study tunes two parameters, ***temperature*** and **top-*p***, to balance the determinism and variability of LLM outputs in knowledge extraction. A higher ***temperature*** setting promotes more diverse outputs, whereas a lower one yields more deterministic results (Renze, 2024). The **top-*p*** parameter constrains the token sampling to high-probability candidate tokens, thereby enhancing output stability and stylistic consistency (Holtzman et al., 2019). Detailed definitions of these two parameters are provided in Appendix C.

3.4.5. Embedding

The purpose of embeddings is to provide a continuous and

computable representation of the extracted knowledge, enabling quantitative operations that cannot be directly performed on symbolic graph elements. Textual contents are converted into semantic vectors by an embedding LLM (referred to as KG encoder) and stored within the database. As a crucial component of the developed KG, the embeddings enable its quality assessment and other downstream applications. In KG evaluation, the embeddings are applied to identify entity redundancy in complexity assessment and quantify semantic fidelity of extracted triplets in accuracy evaluation. Beyond quality assessment, they also facilitate semantic retrieval in knowledge-based RAG systems, allowing the identification of conceptually related information that may not be directly linked in the graph structure.

3.5. Knowledge graph quality assessment

To ensure the reliability and utility of KG generated by LLMs, a comprehensive quality assessment framework is proposed. The KG is assessed in dual aspects: complexity and accuracy. Complexity evaluates the degree of conciseness and structural efficiency of the KG, assessing whether entities, relations, and overall graph structure are represented in a minimal but sufficient form to enclose the underlying knowledge. Accuracy, on the other hand, measures the fidelity of the KG to the original text, evaluating whether the graph correctly captures the key information expressed in the source material. These dimensions reflect a trade-off where reduced complexity supports computational efficiency and interpretability in downstream tasks while excessive simplification may lead to the loss of critical information. Accordingly, quantitative indices are developed to guide the optimization of KG construction.

3.5.1. Complexity assessment

The complexity is primarily assessed through the redundancy of extracted entities, relations, and the graph structure.

(1) Entity Redundancy Rate

Entity redundancy is defined as the presence of multiple entities in the KG that describe the same or highly similar instances (e.g., “strong winds” and “high winds”) leading to unnecessary duplication in the graph representation. Despite the implementation of entity standardization, some semantically equivalent entities may still occur due to the linguistic variability in accident reports. To assess the degree of entity redundancy, a clustering method is employed to group potentially equivalent entities as presented in Appendix D. The entity redundancy rate is defined as the proportion of entities that belong to multi-member clusters of equivalence.

(2) Relation Redundancy Rate

Two types of relation redundancy are considered, which are multiple relations between the same entity pair and self-loops where the head and tails of SPO triplets are identical. These relations violate the schema constraints and reduce the graph clarity. The relation redundancy rate is calculated as the ratio of redundant relations to the total number of relations.

(3) Graph Structure Complexity

To assess whether the developed graph contains redundant or unnecessarily complex structures representing subordinate information in marine accident reports, a graph-entropy measure is introduced. Graph information entropy is a quantitative measure of the structural disorder and information dispersion within a graph, derived from the probability distribution of node degrees or connection strengths. A higher entropy value indicates a more complex graph structure reflecting a greater dispersion of information across nodes and relations. In this study, a main causal chain is first defined, encompassing essential relations that characterize the causation of weather-related marine accidents (e.g., “*causes*”, “*exhibitedResponse*”, “*engagedInOperation*”). Then, the graph information entropy of the main causal chain is compared with that of the corresponding accident-specific KG sub-graph to evaluate whether the structural complexity required to describe the main causal chain dominates the overall sub-graph. The mathematical definition of graph information entropy is presented in Appendix E.

3.5.2. Accuracy assessment

It is a significant challenge to evaluate the semantic faithfulness of the KG against the original text due to the lack of ground truth and the inherent ambiguity in language. In this study, an LLM-based text reconstruction method is proposed to address this issue.

The extracted SPO triplets of each marine accident report are incorporated into a prompt and fed into a generative LLM (the text reconstructor) to reconstruct a complete narrative of the accident

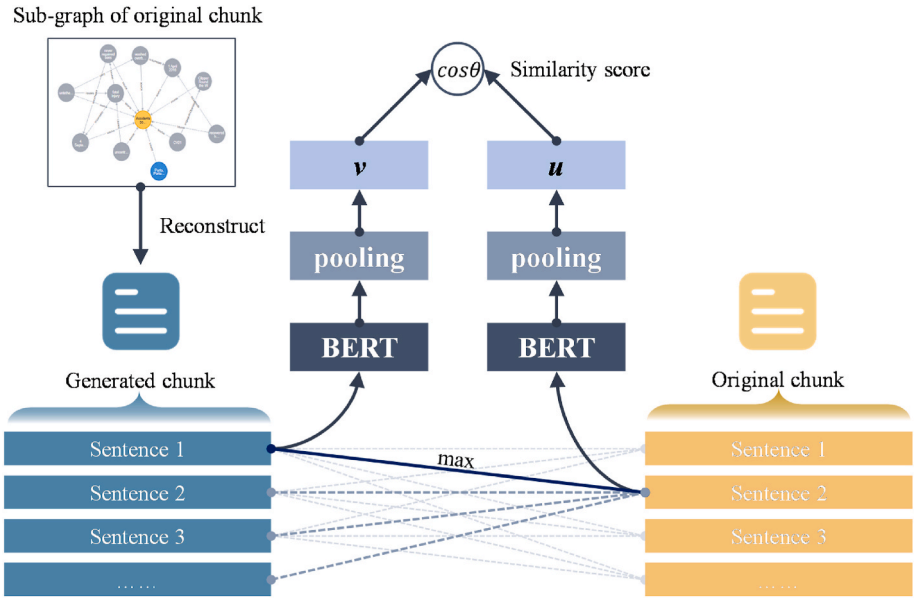


Fig. 2. Sentence-level similarity assessment between generated chunk and original chunk.

Table 2
Detailed information of MAIB reports and MSA reports employed in this study.

Data source	Language	Number of reports after preprocessing	Time span	Selected content	Maximum chunk length
MAIB	English	172	1990–2023	Accident summary, safety issues	730 words
MSA	Chinese	33	2016–2024	Accident description, causation	1800 characters

process. An example prompt for text reconstruction is presented in Appendix B.

To accurately assess the semantic alignment of generated and original chunks, a sentence-level similarity assessment approach is conducted (Fig. 2). Both chunks are segmented into sentences. For each sentence in the original chunk, the highest similarity score with any sentence in the generated chunk is recorded. The average value of these maximum scores is adopted as the ultimate similarity measure. This approach mitigates the influence of word ordering, syntactic variation, and structural mismatch, especially in scenarios where the reconstructed text reorganizes content without changing its core meaning.

A bi-encoder model is applied to provide more robust sentence-level embeddings in the similarity assessment as presented in Fig. 2. In this framework, two input sentences are encoded independently using a shared BERT encoder. Each sentence embedding is obtained by applying a pooling operation over the token embeddings from the final BERT layer. The resulting fixed-size vectors can then be compared using cosine similarity to estimate semantic closeness (Reimers and Gurevych, 2019). Compared to cross-encoder models, bi-encoder models are less affected by reordering or paraphrasing, making them better suited for this evaluation setting, where semantic inclusion and coverage are the primary goals (Thakur et al., 2020).

4. Knowledge graph construction and assessment

4.1. Marine accident investigation report data source

This study utilizes weather-related marine accident investigation reports collected from two authoritative sources: the MAIB in the U.K. and the MSA in China. The focus on weather-related cases is intentionally chosen to maintain the contextual consistency across samples and to emphasize the environmental dimension of accident causation. A detailed justification for this dataset scope has been provided in Section 3.2.

(1) MAIB reports. The MAIB reports marine accidents involving UK-registered vessels worldwide and all vessels operating in the UK territorial waters. From a total of 1067 reports, 172 weather-related cases from 1990 to 2023 are selected via keywords such as “weather”, “heavy weather”, and “rough sea”. The MAIB provides detailed investigation report in PDF format as well as web-source summary of key findings and safety issues. To accommodate the input length constraints of LLMs and minimize preprocessing complexity, the more concentrated web-source information is extracted by web scraping techniques and used to develop the MAIB corpus.
(2) China MSA reports. The China MSA reports marine accidents occurring in both coastal and inland waters in China. Among the officially defined categories, reports in “strong wind-induced accidents” from 2016 to 2024 are selected for this case. All 33 investigation reports in this category are included. As MSA reports are only available in detailed investigation files, relevant sections of accident descriptions and causation analysis are manually extracted to form the second corpus.

Two corpora are processed independently and fed into the KG construction pipeline to verify the robustness and generalizability of the framework across different reporting styles and languages. Summary statistics of the two datasets are provided in Table 2.

4.2. Weather-related marine accident knowledge graph construction

The schema described in Appendix A and the example prompts in Appendix B are applied to the MAIB corpus for knowledge extraction. To ensure cross-language consistency, both the English and Chinese corpora are processed under a unified schema and prompt framework. The schema and prompts for the Chinese corpus are directly translated from the English templates and reviewed by bilingual experts. All entity categories, relations, attribute definitions, instruction format, and output structures are maintained identical cross the two languages.

All LLM services are accessed through API calls to Siliconflow, a commercial AI cloud service platform. To ensure the isolated memory contexts and optimize performance, separate generative models are used for knowledge extraction and text reconstruction. A lightweight, non-reasoning model with fast response time and stable output is utilized as KG extractor. In contrast, a reasoning-augmented LLM is used for text reconstruction where its logical inference capabilities are essential for producing plausible content from unordered triplets. The LLMs used in this case along with the default decoding parameters setting is summarized in Table 3. These generative LLMs and embedding models used are all designed for multilingual and cross-lingual applications. The same models are employed for knowledge extraction across both the English and Chinese corpora (Team Qwen, 2024; Guo et al., 2025; Chen et al., 2024).

From the MAIB corpus, 3612 entities and 9783 relations are extracted. 1263 entities and 2463 relations are obtained from the MSA corpus. The resulting SPO triplets along with their embeddings and the corresponding original text chunks, are written in a Neo4j graph database. To support downstream semantic similarity analysis, the vector embeddings are stored as attributes of entities and relations within Neo4j. This approach enables direct use of embeddings for in-graph similarity computations without the need for an external vector database or plugin. It also simplifies data management and maintains consistency across symbolic and embedding-based representations. A subgraph visualization based on the MAIB corpus is illustrated in Fig. 3. In the graph structure, each accident-level text chunk is incorporated into the graph as a dedicated “chunk” node. All entities extracted from the same chunk are linked to it via a “source” relation (i.e., <entity,

Table 3
LLMs and the default parameter setting in this study.

Role	Model	Parameter size	Context length	Default parameter setting
KG extractor	Qwen2.5-7B-Instruct	7 billion	32k	Temperature: 0.1, Top-*p*: 1.0
Text reconstructor	DeepSeek-R1-Distill-Qwen-7B	7 billion	128k	Temperature: 0.1, Top-*p*: 1.0
Embedding	bge-m3	0.6 billion	8 k	N.A.

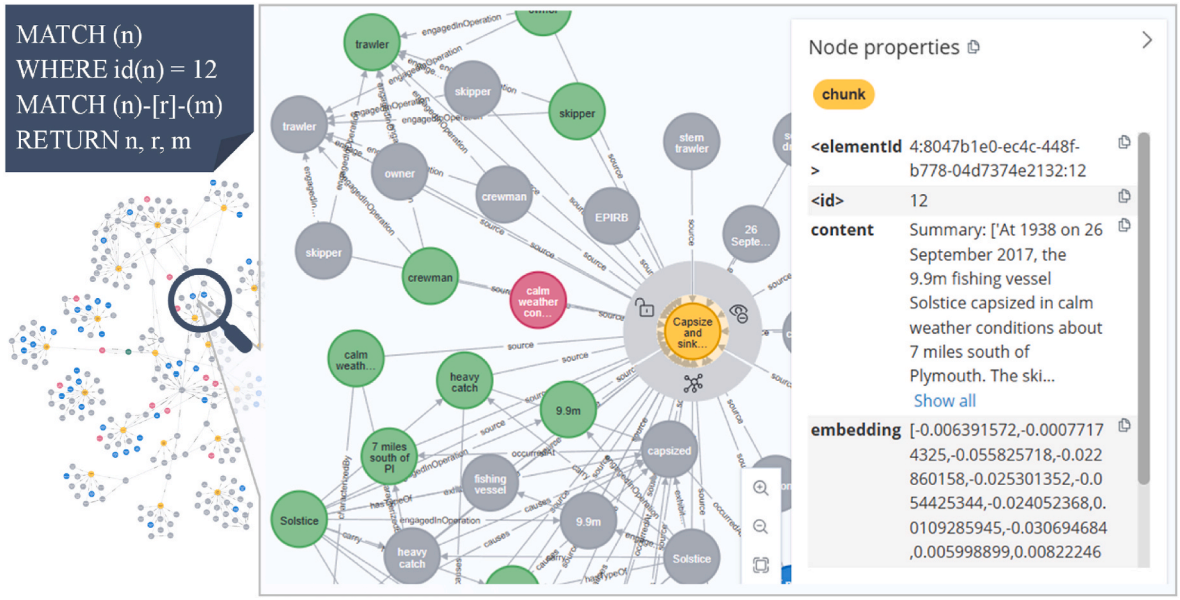


Fig. 3. Sub-graph of MAIB corpus and detailed information of the chunk node retrieved by Cypher.

Table 4
Evaluation results of BERT-based method and LLM-based method on test set.

Metrics type	Metrics	BERT-based method	LLM-based method (Improvement)
Traditional metrics	Macro-average precision	0.7627	0.7856 (+3.0 %)
	Macro-average recall	0.6566	0.7786 (+18.5 %)
	Macro-average F1 score	0.6881	0.7671 (+11.4 %)
Proposed metrics (Complexity)	Entity redundancy rate	0.1200	0.0500 (-58.3 %)
	Relation redundancy rate	0.0000	0.0100 (N.A.)
	Graph structure complexity	0.6611	0.8057 (+21.8 %)
Proposed metrics (Accuracy)	Reconstruction accuracy (Mean)	0.6034	0.6768 (+12.1 %)
	Reconstruction accuracy (Max)	0.8915	0.9243 (+3.6 %)
	Reconstruction accuracy (Min)	0.1357	0.2401 (+76.9 %)
	Reconstruction accuracy (Std)	0.2251	0.1544 (-31.4 %)

source, chunk>) preserving the contextual integrity of the sub-graphs. The detailed information associated with nodes, including contents, vector embeddings, and enumerated attributes, can be retrieved through Cypher queries.

4.3. Comparative analysis of model performance and metric effectiveness

To validate the feasibility of the LLM-based pipeline and the effectiveness of the customized modeling metrics, a comparative analysis is conducted between the LLM-based method and widely employed natural language processing methods (e.g., BERT-based methods), with both established measures and the proposed indicators.

The comparative analysis is exclusively carried out on the MAIB corpus since the sample size of the MSA dataset is insufficient for reliable model training and validation. Entities and relations are annotated by an advanced LLM (ChatGPT 5) prompted via the same schema described in Appendix A. Two BERT-based models, namely BERT-NER model and BERT-based relation classification model, are trained on 70 % of the dataset, with 10 % reserved for hyperparameter tuning and 20 % for testing (Kenton and Toutanova, 2019). Table 4 summaries the evaluation results of BERT-based and LLM-based approaches on test set, respectively. Both traditional metrics and the newly proposed metrics consistently demonstrate the superior performance of applying integrating LLM into schema-based knowledge extraction tasks. Specifically, the substantially higher recall can be attributed to the LLM's stronger contextual understanding which promotes the identification of valid triplets. Furthermore, the enhanced performance in text reconstruction accuracy indicates that the LLM-based method can preserve the semantic coherence of the original text more effectively.

This comparative evaluation also highlights the alignment and difference between traditional metrics and the proposed evaluation indicators. While precision, recall, and F1 evaluates the fidelity of the extracted triplets with respect to the original text, the proposed reconstruction accuracy serves a complementary role from a global semantic perspective. The consistent performance where the LLM-based method outperforms the BERT-based method by roughly 10 % across these two types of metrics, i.e., macro-average F1 score and mean reconstruction accuracy, indicates the validity of the proposed accuracy metrics as a complementary indicator for assessing extraction quality without relying on manually annotated data. Besides, the proposed accuracy metrics measure the global semantic integrity of the extracted knowledge rather than solely on triplet-level correctness. Therefore, it is inherently more sensitive to errors since a single incorrect or missing triplet can disrupt the logical consistency or semantic coherence of the whole reconstructed text. In contrast, the complexity metrics evaluate the structural aspects of the developed KG beyond mere extraction accuracy. Although BERT-based approach obtains satisfactory extraction accuracy in traditional indicators, the developed KG may still exhibit redundant structures and limited interpretability, as reflected by the complexity metrics. The noticeable differences in complexity metrics between two methods highlight the necessity of incorporating structural evaluation.

4.4. Quality assessment of weather-related marine accident knowledge graph

In this section, the LLM-aided KG construction using the default configuration, is evaluated in detail through the proposed evaluation framework. The assessment is performed on both the MAIB and MSA corpora, and the corresponding evaluation results are summarized in Table 5.

Overall, entity redundancy is prevalent in both KGs, whereas the relation redundancy and graph structure complexity remain acceptable. The accuracy indicators suggest that the sub-graphs successfully capture the core semantics of the original accident investigation reports. However, the KG developed from the MSA corpus exhibits lower semantic fidelity compared to MAIB corpus, likely due to variations in language structure and narrative style across the two data sources.

A detailed investigation is conducted to explore the errors produced by LLMs during knowledge extraction. All the redundant clusters with a size larger than 3, as well as all the redundant relations, are recorded for investigation. Through a thorough review of the clustered entities and triplets, typical redundancy types for entities and relations are identified and summarized in Table 6 along with the proportions of each type. In English corpus, morphological variants are the main source of redundant entities, while such issues are absent in the Chinese text. Other redundant entities in the MAIB-based KG mainly stem from inconsistent and vague entity descriptions. In contrast, in the MSA-based KG, synonyms and paraphrased expressions are more common than especially in event descriptions containing longer and more complex phrases. Across both corpora, multiple relations occur in scenarios such as confusion between ship responses and operations, and misinterpretation of schema-defined temporal (*occurredOn*) and spatial (*occurredAt*) relations. Additionally, frequent occurrence of self-loops in both corpora indicates that LLM hallucinations remains a major source of error in relation extraction.

To analyze the source of inaccuracy in more depth, 23 text reconstruction samples with low similarity scores (reconstruction accuracy lower than **Mean - Std.**) are selected from the corpora for further manual review. As shown in Table 7, errors of reconstructed text include (1) introduction of incorrect details or emphasis (e.g., accident causes and consequences): not present in the original text; (2) omission of key intermediate events in the accident process, such as communication or operational responses; (3) oversimplification, such as compressing temporal sequences into single events. High proportion of all errors indicate that multiple error types occur in each sample, which reveals the interdependency of errors and a compounded propagation semantic loss from knowledge extraction to reconstruction. Among these error types, hallucination is the most common error in low-score reconstructed samples. Information sparsity in triplets which omits temporal and causal cues may force the reconstructor to reply on its prior knowledge and fabricate the story. Moreover, these findings also confirm the added diagnostic value of the proposed reconstruction-based approach, which complements traditional methods by exposing the semantic inconsistency.

Table 5
Quality assessment summary for KG based on MAIB and MSA corpus.

Evaluation of Complexity				
Basis corpus	Entity redundancy rate	Relation redundancy rate	Graph structure complexity	
MAIB	0.09	0.01	0.8343	
MSA	0.15	0.01	0.936	
Evaluation of Accuracy				
Basis corpus	Mean	Max.	Min.	Std.
MAIB	0.7607	0.9452	0.3724	0.0992
MSA	0.7071	0.7862	0.2080	0.0944

5. Discussion on theoretical contributions and applications

This section explores how key configuration choices—including schema design, prompt engineering, and LLM decoding parameters—affect the quality of the generated weather-related marine accident KG. Drawing on experimental results, it further discusses strategies for optimizing LLM-based KG construction to enhance both accuracy and efficiency.

5.1. Impact of LLM parameter tuning on KG quality

5.1.1. Accuracy assessment under varying LLM parameters

The results of the accuracy evaluation demonstrate that LLMs with different decoding configurations generally produce KGs of consistent quality. In Fig. 4, Each boxplot represents the similarity score distribution for a specific parameter setting, with the maximum and minimum values indicated. The color intensity of each box reflects the mean values. Dark borders highlight configurations where the LLM generates invalid content requiring post-processing. The reconstructed texts

Table 6
Statistics of typical redundant entities and relations in default configuration.

Basis corpus	Indicators	Redundancy taxonomy	No. of Clusters or Triplets	Entities or relations/ Total redundancy (%)	Examples (in English)
MAIB	Redundant entities	Morphological variants arising from singular/plural and verb variations.	11	38.46	'crew member', 'crew members'
		Compositional or attribute variants sharing the same core concept.	6	27.88	'wooden fishing vessel', 'small fishing vessel', 'fishing vessel'
		Synonyms and paraphrased expressions.	6	25.01	'rough weather', 'adverse weather', 'poor weather'
		Format variants.	3	8.65	'Mayday', 'MAYDAY'
	Redundant relations	Multiple relations	10	41.7	head: sightseeing excursion, tail: October 18, 2004, relations: ['occurredOn', 'occurredAt']
		Self-loops	14	58.3	head & tail: heavy weather, relation: characterizedBy
MSA	Redundant entities	Synonyms and paraphrased expressions.	5	40.98	'Inadequate typhoon response', 'Typhoon preparedness was lacking'
		Compositional or attribute variants sharing the same core concept.	4	29.52	'Water ingress into the engine room', 'Seawater entered the cargo hold'
		Format variants.	3	29.50	'force 15 gale', 'force 15 wind'
	Redundant relations	Multiple relations	1	16.7	head: Shipname, tail: anchor dragging, relations: ['engagedInOperation', 'exhibitedResponse']
		Self-loops	5	83.3	head & tail: Timestamp, relation: occurredAt

Table 7
Review of reconstructed text errors based on low-similarity score samples.

Error category	Potential cause	Proportion (% of all the selected cases)	Example case		
			Original text	SPO triples	Reconstructed text
Hallucination and factual distortion	The KG extractor fails to extract valid SPO triplets for describing the facts.	86.95	Three people injured: two with bruises, one with a head injury (skipper).	<Three people (casualty), *source*, chunk>	Two with puncture wounds, one with a broken leg.
Loss of situational details	Incomplete triplet extraction.	82.60	Describes the skipper's route, reliance on GPS, unease, speed reduction, and the grounding.	<Wizard (shipname), *exhibitedResponse*, slamming>	A general statement of slamming.
Event compression and temporal flattening	Caused by sparse or vague triplet input.	60.87	The vessel left the quay at 0520. The skipper increased speed at 0524. Grounding occurred at 0528.	<0520 (timestamp), *source*, chunk> <0524 (timestamp), *source*, chunk> <0528 (timestamp), *source*, chunk>	Incident occur at 0520.

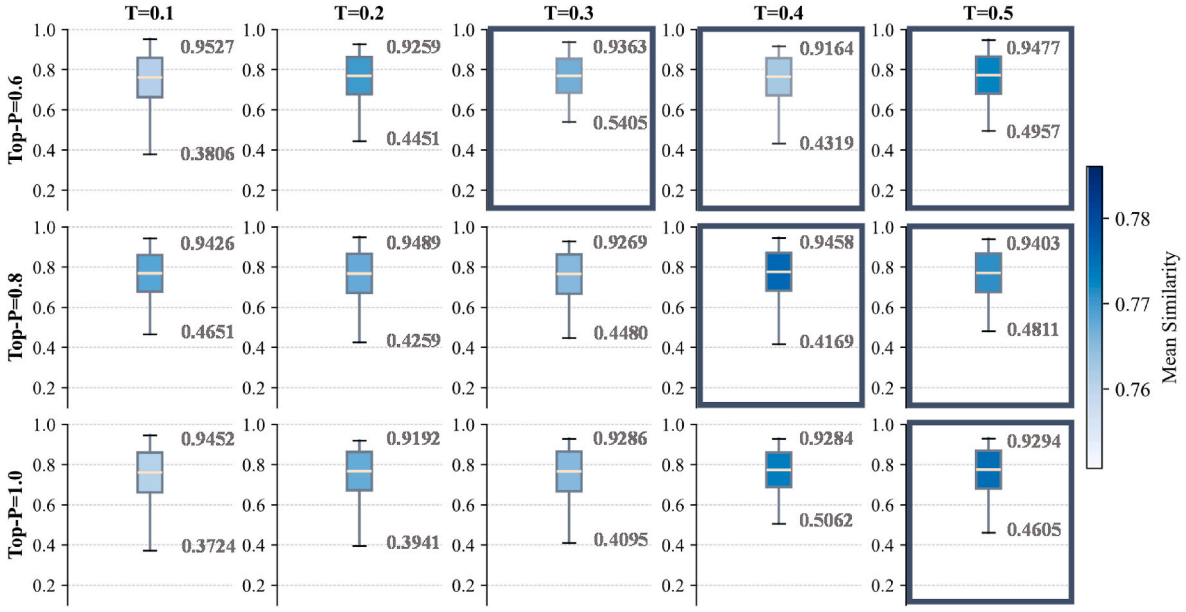


Fig. 4. Fidelity score distributions of reconstructed text chunks under varying LLM decoding parameters (temperature, denoted as T, and Top-*p*) on the MAIB corpus.

Table 8
An example of comparison among original chunk and reconstructed chunks under varying LLM parameters.

Source	Core context
Original chunk	On August 4, 2018, the UK registered container vessel, ANL Wyong, and the Italian registered gas carrier, King Arthur, collided in darkness, dense fog and an area of heavy shipping traffic, south-east of Gibraltar; both vessels were destined for Algeciras, Spain. ANL Wyong was stopped and waiting for a pilot as there was no berth immediately available; King Arthur was heading towards a boat transfer position near Algeciras. The accident happened because neither vessel appreciated the risk of collision in sufficient time to take effective avoiding action and pass at a safe distance. The investigation has also highlighted risks associated with the inappropriate use of VHF radio and AIS information when assessing risk of collision. The collision occurred within a recognized vessel traffic service (VTS) area; however, the vessels were not warned of the developing risk by the shore authority responsible for traffic safety in the area.
Reconstructed chunk (temperature = 0.1, Top-p = 0.6)	*On August 4,2018*, the *container vessel ANL Wyong* encountered severe weather conditions characterized by *darkness and dense fog* while operating in the *VTS area*. The ship's master had previously **planned to alter the passage** to avoid these conditions. However, due to the unpredictable nature of the weather, the crew continued their operations in heavy darkness and fog, leading to operational challenges. During this period, AIS information and VHF radio communications were exchanged, which were part of a **bad emergency response**. The officer of the watch **reported manning** issues, while the master's action was identified as a contributing factor. These factors, combined with the adverse weather conditions, resulted in a collision involving the ANL Wyong and another vessel.
Reconstructed chunk (temperature = 0.5, Top-p = 0.6)	*On August 4,2018*, two ships, *ANL Wyong (a container vessel)* and *King Arthur (a gas carrier)*, were sailing **between** *Algeciras, Spain, and south-east of Gibraltar*. The situation became severe when *darkness and dense fog* characterized the weather, which **was characterized by** the operation of the *VHF radio and AIS information*. Unfortunately, no action was taken on board ANL Wyong, while King Arthur's master's actions were deemed insufficient. The *lack of action* and misleading AIS information, which was further complicated by the VHF radio, led to a collision occurring within a designated *VTS area*. Despite the collision, there were no casualties or injuries reported.

Italic contents with underlines indicate true elements present in the original chunk.
Contents in bold format indicate the fabricated and incorrect elements.

achieve average similarity scores of about 0.77 across all the configurations, indicating that most of the core information has been preserved by the KG extractor. Despite this overall stability, variations in temperature and Top-*p* settings do influence specific aspects of extraction performance. Notably, higher temperature values are associated with greater semantic fidelity in reconstructed texts. Contrary to expectations, the results indicate that lower temperature settings do not always lead to more accurate or schema-aligned outputs. Moreover, configurations involving low Top-*p* and high temperature tend to generate more invalid content including entities lacking type annotations, incomplete relations, and JSON format violations. These outputs require post-processing steps, such as deletion and format repairs prior to quality evaluation.

To better understand the above results, a detailed comparative analysis is conducted based on randomly selected samples. Table 8 presents an example illustrating the differences under varying LLM decoding settings. Under lower temperature setting (e.g., 0.1), the model adheres more rigidly to the schema, occasionally inserting predefined entities or relations not explicitly mentioned in the text. These outputs may appear as hallucinations but are schema-induced errors. By contrast, a higher temperature setting (e.g., 0.5) enables the model to match the entities and relation in a more flexible manner despite the constrains of schema. Especially when the schema does not fully cover the scope of targeted ontology embedded in the corpus, the LLM with higher temperature extracts more contextually grounded or marginal information. This suggests a trade-off between schema compliance and textual fidelity that is mediated by the temperature parameter.

5.1.2. Complexity assessment under varying LLM parameters

Fig. 5 illustrates the complexity assessment results under different configurations on MAIB corpus. In the subfigure (a), bubble size indicates the total number of extracted entities, with values labeled, and color intensity reflects entity redundancy. Subfigure (b) presents relation counts using square size, with color intensity indicating relation redundancy. Subfigure (c) applies color intensity to visualize structural complexity for each parameter setting. As presented in Fig. 5, the complexity evaluation indicates that entity redundancy rates remain below 22.5 %, relation redundancy rates are under 3 %, and the main chains account for approximately 80 % of the total information content in the graph. These findings suggest that the LLM produces relatively stable redundancy levels across varying configurations. Nevertheless, decoding configurations, such as temperature and Top-*p*, have potential impacts on the complexity of developed KG. Generally, higher Top-*p* values (e.g., $P = 1.0$) can reduce the duplicated entities and relations. Moreover, a combination of low temperature ($T \leq 0.2$) and low Top-*p* ($P = 0.6$) tend to yield more concise graph structures, focusing on core information.

To further assess redundancy, the most frequent redundant entity types within the potential equivalent clusters and invalid relations were analyzed. These types remain consistent across all decoding parameters and match those identified under the default setting (see Table 5). Notably, as Fig. 5 presents, configurations with higher temperature results in fewer total extracted entities and relations overall. This outcome is probably due to: (1) the removal of invalid outputs during post-processing, and (2) the LLM's inclination to prioritize salient and representative information when not constrained by strict formatting requirements. Consequently, changes in redundancy metrics are primarily attributed to variations in the total number of extracted entities and relations rather than improvements in the standardization process. It underscores the importance of enhancing standardization to reduce redundancy more effectively, as parameter tunning cannot ensure consistent outputs.

Furthermore, low temperature and low top-p settings tend to produce graph structures in which the main chains concentrate a larger share of the overall information content. This aligns with the expected behavior of LLMs under such constrained decoding configuration. The model tends to extract more confident and semantically central SPO triplets, such as ship responses, equipment failures, and accident consequences. These elements usually compose the causal chains of the marine accident narratives, characterized by more explicit sentence patterns and more frequent terminology, thereby increasing their likelihood of being consistently captured.

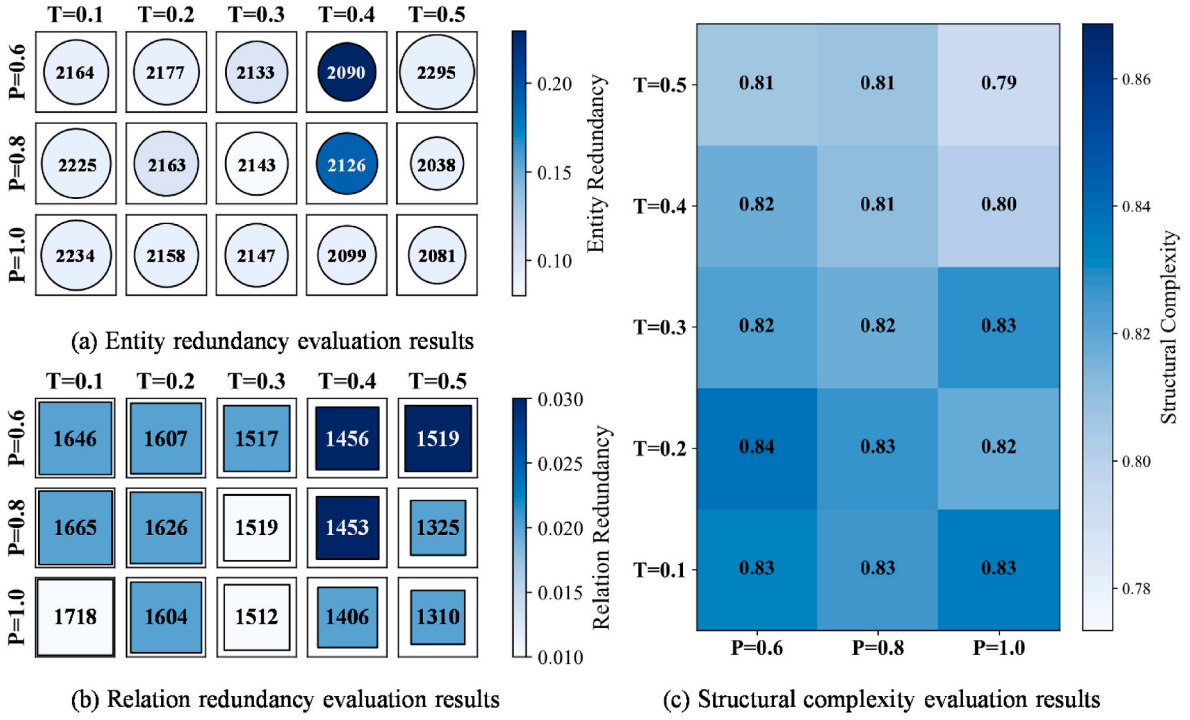
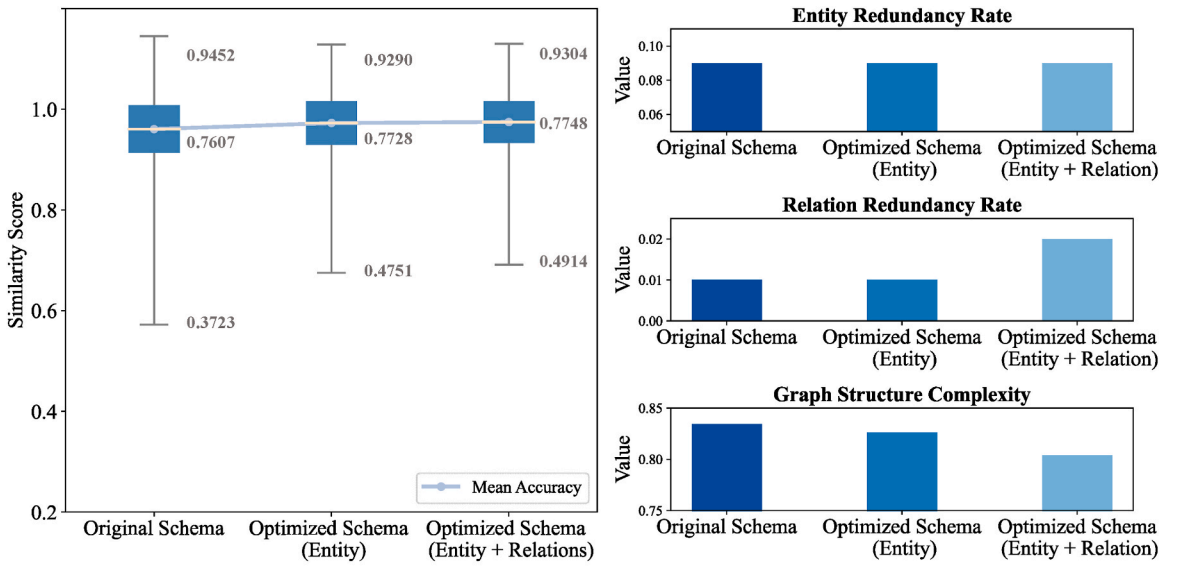


Fig. 5. Complexity evaluation of entities, relations, and graph structure under varying LLM decoding parameters (temperature, denoted as T, and Top-*p*, denoted as P) on the MAIB corpus.



(a) Accuracy evaluation results under different schemas (b) Complexity evaluation results under different schemas

Fig. 6. KG Quality evaluation across varying schema designs: original schema, schema with optimized entity type definitions, and schema with optimized relation type definitions.

5.2. *Impact of schema and prompt design on KG quality*

This section discusses the impact of schema and prompt design on the quality of generated KG for marine accident analysis. To systematically assess their effects, schema and prompt revisions are implemented in a step-wise manner, with each step introducing incremental modifications. This design allows for the exploration of KG quality sensitivity to specific configuration changes and enables the identification of key factors that contribute to KG quality improvements.

5.2.1. *KG quality assessment under incremental schema optimizations*

The schema is revised by introducing more fine-grained entity and relation types to improve extraction precision and semantic clarity. In the first phase, broad entity categories, such as *ContributingFactor*, are subdivided into specific types according to their enumerated attributes, e.g., *EquipmentFailure*, *HumanErrors*, *MaintenanceIssue*. Additionally, descriptions are incorporated to guide the KG extractor in accurately aligning extracted entities with schema-defined concepts. In the second phase, the generic *cause* relation is disaggregated into more nuanced types, including *directCauseOf*, *exacerbates*, and *resultsIn* for better distinction between primary, secondary, and compounding causal links in accident narratives. All experiments are conducted under the default KG extractor parameter setting (temperature is 0.1 and Top-*p* is 1.0).

Fig. 6 illustrates the results of accuracy and complexity evaluation under incremental schema optimizations. Fig. 6(a) indicates that the schema refinement improves the overall semantic fidelity of the extracted knowledge SPO triplets. Notably, the subdivision of entity types contributes more to similarity score improvements than modifying relation schema. These improvements are particularly significant in cases where semantically central information is previously missed resulting in low similarity between original and reconstructed texts. The gains in accuracy are attributed to the increased semantic granularity and its constraint effect, which collectively enhance entity alignment, reduce ambiguity, and guide the LLM to a narrower and more precise output search space.

Fig. 6(b) shows that while overall KG accuracy gets improved, relation redundancy increases and the graph becomes structurally more complex, incorporating additional branches to represent subordinate content. Refining the entity schema into finer-grained types improves the accuracy of the NER sub-task by helping the model better align mentions with their corresponding entity types, reducing errors caused by overly broad schema. In contrast, refining relation types does not yield the desired effect and instead introduces semantic ambiguity and overlap between relations. For example, the relations *exacerbates* and *causes* are both assigned between "pitching" and "container stow collapse". While *causes* typically indicates that an event leads to the emergence of a new problem, *exacerbates* implies the intensification of an already existing issue. The model, however, fails to fully distinguish this nuance when presented with a more fine-grained relation schema, resulting in redundant and overlapping relation assignments. Moreover, invalid triplets, such as < crew member died, *resultsIn*, crew member died>, highlight the model's inability to detect or prevent self-loops. These results suggest that while schema refinement enhances expressiveness and accuracy, it may also lead to redundant or overlapping relation assignments, especially when contextual or structural constraints (e.g., mutual exclusivity or relation validity) are not explicitly enforced.

5.2.2. *KG quality assessment under incremental prompt optimizations*

Prompt designs are refined at each stage of the knowledge extraction pipeline to improve the KG quality. In the NER step, the prompt is enhanced with additional instructions emphasizing the completeness of entity detection, clarity and conciseness in naming, and deduplication of semantically identical entities. For entity standardization, illustrative examples are incorporated into the prompt to demonstrate how to unify similar terms into consistent expressions, thereby mitigating the entity redundancy. In the relation extraction step, targeted prompt revisions are introduced to reduce relation redundancy. Specifically, additional prompts are incorporated to help the model identify and avoid two common types of errors: (1) assigning multiple spurious relations to the same pair of entities, and (2) generating self-loop triplets where the subject and object are the same entity. Table 9 summarizes the additional prompt instructions applied for each stage of the extraction process.

The evaluation results presented in Fig. 7 reveal trade-offs between semantic accuracy, redundancy control, and core structure preservation. Fig. 7(a) shows the evident effects of prompt enhancements on improving the overall accuracy of the KG. However, the revised relation extraction prompt does not further increase the semantic fidelity. One possible explanation is that such instructions of self-checking

Table 9
Prompt revisions in each step of knowledge extraction.

Steps in knowledge extraction	Additional instructions added into the prompt
Named entity recognition	1. Carefully inspect the full text to ensure no potential entity is missed. 2. For each entity, provide all relevant attributes defined in the schema if mentioned in the text. 3. Use concise but informative names that accurately reflect the content and follow naming conventions. 4. If an entity appears multiple times in the text, merge them into a single entry and consolidate their attributes.
Entity standardization	1. Unify semantically similar terms using consistent vocabulary (e.g., "poor weather", "rough weather", "heavy weather" → "adverse weather"; "serious damage", "substantial damage" → "major damage"). 2. If the same concept is described using different phrasings, standardize them into a single preferred form based on domain conventions. 3. Avoid vague or overly general names if a more precise term is commonly used in maritime reporting.
Relation extraction	1. Do not generate self-loops: the head and tail of each relation triplet must refer to different entities. 2. Avoid multiple relations with the same head and tail pair: if several candidate relations exist between the same entity pair, choose the most semantically accurate and domain-appropriate one. 3. Do not duplicate any relation. Each relation triplet (subject-predicate-object) must be unique. 4. All extracted relations must conform strictly to the provided schema. Do not infer or fabricate relation types not in the schema. 5. End the output after listing all valid and unique relations.

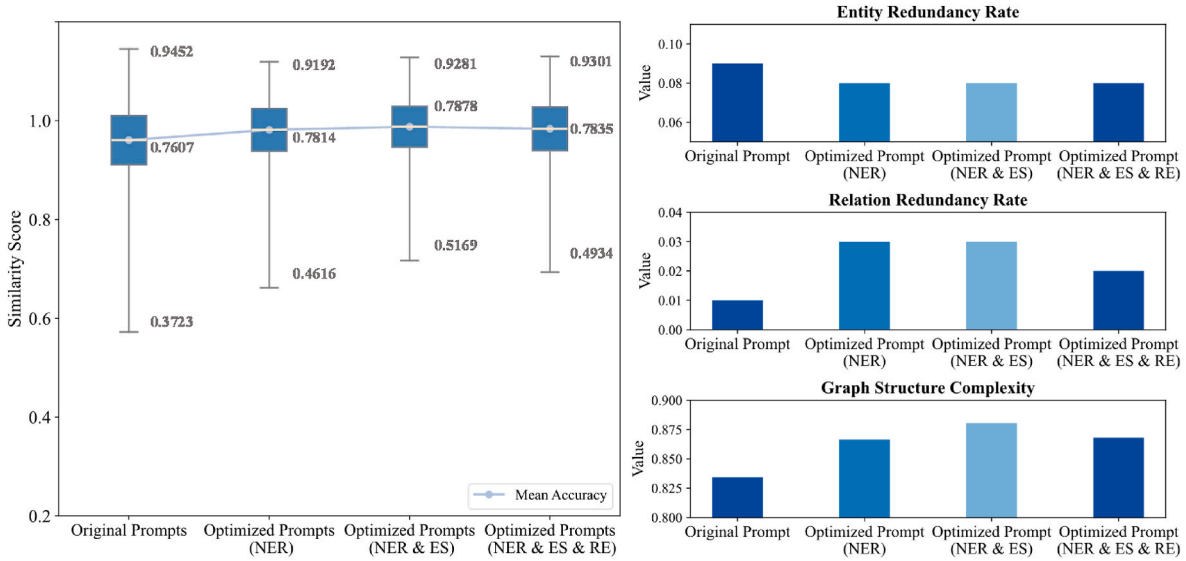
mechanisms introduce additional cognitive load on the model, leading to distraction from primary extraction task (Hwang et al., 2025). Moreover, when redundant relations are encountered, it may discard the triplets including the legitimate ones, thus inadvertently reducing performance. Fig. 7(b) illustrates a complicated and mutually restraining effect of prompt adjustments on KG complexity. First, optimizing NER prompts successfully reduces entity redundancy and enhanced main chain clarity, but increases relation redundancy. The potential reason is that the expanded set of identified entities open more potential pairings for relation extraction. Second, despite the improvements in the prompt for entity standardization, the entity redundancy rate has not been effectively reduced as expected. Manual review of the equivalent entity clusters reveals that only entities with near-identical names are merged, while most of the redundant entities such as mentions of adverse weather conditions and ship types are not unified. Lastly, relation prompt constraints, though effective in reducing spurious and redundant relations, also suppresses some valid and informative main chain links, leading to a measurable decline in structural completeness. These findings underscore the importance of balanced prompt engineering strategies in LLM-based extraction. While prompt refinement improves specific aspects of the developed KG, overly rigid instructions may compromise other dimensions. This highlights the need to explore adaptive prompting mechanisms that dynamically adjust LLM behaviors, particularly in the context of domain-specific tasks.

5.3. *Domain knowledge-enhanced optimization for maritime safety KG construction*

Based on the proposed assessment framework and experimental findings, an iterative optimization process tailored to the characteristics of marine accident KGs is proposed. For each iteration, the process comprises four stages—evaluation, diagnosis, adjustment, and re-evaluation—guided by domain-specific considerations:

(1) Evaluate the developed KG using complexity and accuracy indicators.
(2) Diagnose performance bottlenecks by comparing observed indicator values with predefined targets, such as redundancy in entities (e.g., vessel types and weather conditions), misalignment of hazardous event timelines, and omission of key accident causations.
(3) Refine the schema, prompts or decoding parameters based on diagnostic insights.
(4) Re-construct and re-evaluate the KG quality to assess improvements.

In Step 1, a quantitative assessment is performed using the indicators proposed in Section 3.5 to assess the current quality of the KG. Step 2 determines the optimization objective for the current iteration by



(a) Accuracy evaluation results under different prompt designs

(b) Complexity evaluation results under different prompt designs

Fig. 7. KG Quality evaluation across varying prompt designs: original prompts, optimized prompt for NER, optimized prompt for entity standardization (ES), optimized prompt for relation extraction (RE). In each step, one more prompt is optimized in order.

linking evaluation outcomes with specific refinement strategies. Given the inherent trade-off between preserving accuracy and reducing complexity, each iteration typically focuses on improving one dimension at a time. Step 3, the core of the quality optimization process, involves adjusting various configuration components based on diagnostic insights. The following paragraphs detail specific adjustments tailored to the maritime transport safety domain, with a focus on either reducing the complexity or improving the accuracy.

When the indicators show excessive redundancy or invalid structures, configuration adjustments should aim to streamline the graph. For example, schema refinement can draw on taxonomies according to the marine accident reporting standards by subdividing broad entity types, such as *MaintenanceIssue*, into granular subtypes (e.g., failure not detected, improper repair, inadequate maintenance, and lack of maintenance) each with each with domain-relevant attributes, such as vessel component and status. Entity standardization can be supported by integrating a reference list of canonical marine accident terminology to assist LLMs in merging semantically equivalent entities. During relation extraction, prompts should instruct the model to catch the causal logic common in maritime safety investigations, and to check and filter out invalid SPO triplets. Conservative decoding parameter setting can help control generation variability and reduce noise.

If accuracy is the main concern, optimizations should focus on improving extraction fidelity, particularly around accident causal sequences and timelines. Schema enhancements may include explicitly defining time-sensitive relations like *occurredOn*, *occurredAt*, or cause, ensuring proper alignment of events such as engine failure and emergency response. Slot-filling templates can help ensure all critical accident attributes—such as vessel class, incident location, and casualties—are systematically extracted. Prompt refinements should emphasize the expected content (e.g., causes, outcomes) and include timestamp alignment instructions to ensure accurate temporal linking. Experiment results suggest that a more exploratory parameter configuration may improve the precision of knowledge extraction.

Moreover, additional considerations include the inherent variability of LLM outputs and their sensitivity to schema design. Since LLMs may generate different results across runs, aggregating outputs from multiple generations can yield more stable and robust results. Furthermore, schema sensitivity implies that iterative, fine-grained refinements allow performance impacts to be evaluated incrementally. Analysis of specific low-quality extractions provides valuable feedback for adjusting schema components for specific entity types and relations, enabling a more controlled and resilient optimization process.

Finally, Step 4 involves re-constructing with refined configurations and re-evaluating the KG using the same indicators. This is a critical step for verifying whether the adjustments made in the current iteration effectively address the diagnosed issues. The iterative re-evaluation process ensures steady quality improvements and provides empirical feedback to guide further refinement. Through this closed-loop process, domain knowledge is continuously integrated and validated, enabling the construction of more accurate, compact, and interpretable KGs tailored to marine accident analysis.

5.4. Practical applications and implications

The automated marine KG construction pipeline of marine accident, in conjunction with the evaluation framework, offer practical value for enhancing the data integration, regulatory efficiency, and operational intelligence in maritime safety management.

The LLM-based KG construction pipeline enables maritime authorities to maintain continuously updated knowledge bases. By automatically extracting causal factors and safety violations from newly issued investigation reports, it extensively reduces the reliance on labor-intensive manual review in KG updating. Through transforming large volumes of unstructured marine accident narratives into structured graph model under a predefined schema, quantitative risk assessment on accident reports becomes more feasible. Statistical evaluation of key entities (e.g., weather condition, equipment failures or human factors) and their causal chains over a broader time span can reveal trends of emerging hazards, offering insights for early warning. Moreover, the automated constructed KG enhances knowledge retrieval and reasoning capabilities in operational contexts. By connecting historical accident records with shared nodes, such as vessel characteristics, environmental conditions, and causal mechanisms, a RAG system incorporating such KGs can align ongoing operational states with analogous historical incidents (Ren et al., 2025). This capability enables real-time decisions for crew members and safety officers, assisting in the identification of potential risk escalation pathways and the formulation of appropriate mitigation strategies.

The proposed KG quality evaluation framework is fundamental for more reliable real-world applications of KGs in maritime safety management. The introduced metrics can help identify redundancy and semantical loss in extracted knowledge prior to KG deployment within safety regulatory or accident investigation systems. The framework forms a quantitative feedback loop, enabling continuous optimization of the schema design, prompt formulation, and model decoding parameters setting without the need for extensive manual data annotation. Moreover, this evaluation framework provides a robust foundation for benchmarking different KG construction methods and supports the development of standardized and validated KGs for maritime safety analysis and decision support applications.

6. Conclusion

This study has explored the application of LLMs for automated KG construction and assessment within the maritime safety domain. A knowledge extraction pipeline has been proposed, encompassing NER, entity standardization, and relation extraction, with one-shot prompting techniques employed to guide LLM outputs using a predefined schema. To assess the resulting KG, a dual-perspective evaluation framework is developed measuring the semantic fidelity and structural redundancy. The proposed method is applied to weather-related marine accident investigation reports from both the MAIB in the UK and the MSA in China, demonstrating its adaptability across linguistic and narrative variations. Through systematic experiments, the impacts of LLM decoding parameters, schema, and prompts on the LLM-based knowledge extraction are analyzed. The experiment results reveal inherent trade-offs among semantic accuracy, redundancy control, and core structure preservation. Based on these findings, an iterative optimization framework is proposed, incorporating domain knowledge to refine schema, prompts, and model parameters for improved KG generation in the maritime safety context.

Despite these contributions, several limitations remain. The main issues and corresponding future research directions are summarized as follows. First, while the current knowledge base is limited in weather-related scope, extending the framework to a wider range of accident types remains a crucial future direction. Such an extension will increase the complexity of schema design and highlight the trade-off between schema granularity and the LLM's capability for comprehension. Second, future research could explore the integration of diverse data sources including maritime accident records, deck logs, Automatic Identification System (AIS) data, and communication transcripts. Combining textual, numerical, and sensor-based information may allow the knowledge graph to capture richer representations of vessel behavior and environmental conditions. Moreover, future studies could examine cross-lingual adaptive mechanisms such as multilingual embeddings and schema alignment to improve the transferability and robustness of KG construction in diverse maritime jurisdictions and linguistic contexts.

CRediT authorship contribution statement

Yujie Huang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Ran Yan:** Writing – review & editing, Supervision, Data curation, Conceptualization. **Zhipeng Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT 4o in order to improve the language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX A. Original schema for weather-related marine accident knowledge extraction in JSON format

```
{
  "concepts": [
    { "name": "Vessel", "description": "A ship or boat that is used for maritime operations." },
    { "name": "Weather", "description": "The meteorological conditions at a specific time and place." },
    { "name": "HazardousEvent", "description": "An event that poses a risk to maritime operations." },
    { "name": "Accident", "description": "An unexpected event that results in damage or injury." }
  ],

  "entitytypes": [
    { "name": "ShipName", "subClassOf": "Vessel" },
    { "name": "ShipType", "subClassOf": "Vessel" },
    { "name": "Cargo", "subClassOf": "Vessel" },
    { "name": "ShipOperation", "subClassOf": "Vessel" },
    { "name": "WeatherType", "subClassOf": "Weather" },
    { "name": "WindForce", "subClassOf": "Weather" },
    { "name": "SeaState", "subClassOf": "Weather" },
    { "name": "Visibility", "subClassOf": "Weather" },
    { "name": "Date", "subClassOf": null},
    { "name": "Location", "subClassOf": null },
    { "name": "Timestamp", "subClassOf": null },
    { "name": "ContributingFactor", "subClassOf": "HazardousEvent" },
    { "name": "ShipResponse", "subClassOf": "HazardousEvent" },
    { "name": "AccidentEvent", "subClassOf": "Accident" },
    { "name": "Casualty", "subClassOf": "Accident" },
    { "name": "CargoLoss", "subClassOf": "Accident" }
  ],

  "enumerations": {
    "ShipOperationEnum": [
      "fishing", "sailing", "Alongside/moored/anchored", "maintenance", "other"
    ],
    "ShipResponseEnum": [
      "rolling", "listing", "pitching", "slamming", "cargoshift", "water Ingress", "Vibration", "other"
    ],
    "ContributingFactorEnum": [
      "human factor", "equipment failure", "maintenance issue", "manning issues", "bad workplace conditions",
      "Inadequate tools and equipment", "bad emergency response", "other"
    ],
    "AccidentEventEnum": [
      "collision", "grounding", "contact", "fire", "hull failure", "loss of control", "vessel damage",
      "Capsize/listing", "Flooding/foundering", "Ship missing", "other"
    ]
  },

  "attributes": [
    { "name": "operation_type", "domain": "ShipOperation", "range": "ShipOperationEnum" },
    { "name": "response_type", "domain": "ShipResponse", "range": "ShipResponseEnum" },
    { "name": "factor_type", "domain": "ContributingFactor", "range": "ContributingFactorEnum" },
    { "name": "event_type", "domain": "AccidentEvent", "range": "AccidentEventEnum" }
  ],

  "relations": [
```

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(*continued*)


```
    { "name": "hasTypeOf", "source": "ShipName", "target": "ShipType" },
    { "name": "carry", "source": "ShipName", "target": "Cargo" },
    { "name": "engagedInOperation", "source": "ShipName", "target": "ShipOperation" },
    { "name": "exhibitedResponse", "source": "ShipName", "target": "ShipResponse" },
    { "name": "characterizedBy", "source": "WeatherType", "target": "WindForce" },
    { "name": "characterizedBy", "source": "WeatherType", "target": "SeaState" },
    { "name": "characterizedBy", "source": "WeatherType", "target": "Visibility" },
    { "name": "occurredOn", "source": "ShipOperation", "target": "Date" },
    { "name": "occurredAt", "source": "ShipOperation", "target": "Location" },
    { "name": "occurredAt", "source": "ShipOperation", "target": "Timestamp" },
    { "name": "occurredAt", "source": "ShipResponse", "target": "Timestamp" },
    { "name": "occurredAt", "source": "AccidentEvent", "target": "Timestamp" },
    { "name": "causes", "source": "WeatherType", "target": "ShipResponse" },
    { "name": "causes", "source": "ShipResponse", "target": "ShipResponse" },
    { "name": "causes", "source": "ShipResponse", "target": "CargoLoss"},
    { "name": "causes", "source": "ShipResponse", "target": "AccidentEvent" },
    { "name": "causes", "source": "ContributingFactor", "target": "AccidentEvent" },
    { "name": "causes", "source": "WeatherType", "target": "AccidentEvent" },
    { "name": "causes", "source": "AccidentEvent", "target": "Casualty" }
    { "name": "causes", "source": "AccidentEvent", "target": " CargoLoss " }
  ]
}
```


APPENDIX B

Prompt for named entity recognition:

You are an expert in Named Entity Recognition (NER) in the maritime safety domain.
Based on the schema and example below, extract all entities and their corresponding attributes from the given text chunk according to the schema constraints.
Schema:
Concepts: {concepts}
Entitytypes: {entitytypes}
Enumerations: {enumerations}
Attributes: {attributes}
Example:
Text chunk: {example_text}
Extraction result: {example_output}
Now, please extract information from the following text chunk:
Text Chunk: {text_chunk}
Important Instructions:
Entities:
1. Only extract entities that explicitly appear in the text.
2. The number of entities should be limited to less than 20.
3. Do not duplicate any entity. Each entity should only appear once in the list.
Output format:
1. Please return the result strictly in JSON format.
2. Do not include any explanations, comments, or non-JSON content.
3. Ensure that all JSON keys and string values are enclosed in double quotes (").
4. Make sure that all brackets and braces are correctly matched and the overall JSON is well-formed.
5. If no valid entity can be extracted based on the schema, return an empty JSON array:`[]`.

Prompt for entity standardization:

You are a domain expert in maritime safety and knowledge graph construction. Your task is to standardize the surface forms of named entities extracted from accident reports.
Each entity is represented as a JSON object with the fields: "name", "entitytype", and "attributes". You must only modify the "name" field to its canonical or normalized form while preserving the original meaning.
The following is the designed schema and an example of the input and output format:
Schema:
Concepts: {concepts}
Entitytypes: {entitytypes}
Enumerations: {enumerations}
Attributes: {attributes}
Example:
Text_chunk: {example_text}
Given NER results:{example_ner_results}
Standard NER results: {example_std_ner}
Now, please output the list of updated entity JSON objects with only the "name" fields modified accordingly. Do not include any explanations or additional comments.
Text Chunk: {text_chunk}
Given NER results: {ner_results}
Instructions for standardization:
1. Replace verbose or overly specific names with their standard or commonly accepted equivalents.
2. Use singular form unless the plural has specific domain significance.
3. Use ISO format for dates (e.g., "October 30, 2017" → "2017-10-30″).

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4. Use simplified terminology for repeated variants (e.g., "bow flare slamming" → "slamming", "rough weather" → "adverse weather"). 5. For complex descriptive phrases, reduce them to concise professional terms (e.g., "containers were not stowed … " → "improper stowage"). 6. Do not change the underlying semantics. Instructions for output format: 1. Please return the result strictly in JSON format. 2. Do not include any explanations, comments, or non-JSON content. 3. Ensure that all JSON keys and string values are enclosed in double quotes ("). 4. Make sure that all brackets and braces are correctly matched and the overall JSON is well-formed. 5. If no valid entity can be extracted based on the schema, return an empty JSON array:'[]'. 6. Do not return Python syntax (e.g., None, True, False). Use JSON syntax instead (null, true, false).
Prompt for relation extraction: You are an expert in the maritime safety domain. Based on the schema and example below, extract all relations from the given text chunk according to the schema constraints and extracted entities. Schema: relations: {relation_dict} Example: text chunk: {example_text} extracted entities and attributes: {example_ner_results} relation extraction result: {example_output} Please extract relation triplets from the given text chunk based on the provided list of extracted entities and attributes. text chunk: {text_chunk} list of extracted entities and attributes: {ner_results} Important Instructions: 1.Return the results strictly in JSON format. 2.Do not include any explanations, comments, or non-JSON content. 3.Ensure that all keys and all string values are enclosed in double quotes ("). 4.Make sure all curly braces and square brackets are correctly matched and balanced. 5.If no valid relation can be extracted based on the schema, return an empty JSON array:'[]'. 6.Do not duplicate any relations. Each relation should only appear once in the list. 7.End the output after listing all unique relations.
Prompt for text reconstruction: You are a language generation assistant in maritime safety domain. Your task is to reconstruct a natural, coherent, and semantically faithful descriptive text of marine accident summaries and causations based on the structured information of entities and relations. Below is an example of a reconstructed text: Entity example: {entity example} Relation example: {relation example} Reconstructed text example: {reconstructed text example} Please follow these instructions carefully: 1.Do not introduce any new information beyond the input. 2.Use only the exact entities and relations listed. 3.Do not infer or guess missing facts. If uncertain, omit. 4.Use natural language, but ensure every sentence is directly supported by at least one entity or relation. 5.Reuse the wording of entities where possible. 6.The language should be fluent and coherent. Here is the structured information: Entities: {entities} Relations: {relations} Now, please generate a natural language description based on the above information.

APPENDIX C

The decoding parameter ***temperature* (*T*)** adjusts the probability distribution of output tokens. It is introduced to reassign the probabilities of candidate tokens as follows:

$$\widehat{p}_i = \frac{\exp\left(p_i/T\right)}{\sum_j \exp\left(p_j/T\right)} \tag{C-1}$$

where $\widehat{p}_i$ is the adjusted probability of token i; p_i is the original probability of token i; T is the temperature. Low temperature value setting can enable the LLM to produce more deterministic and focused outputs. However, excessively low temperatures may increase the risk of repetitive or trapped generation.

The **top-*p*** parameter, also known as nucleus sampling, limits the sampling space during the generation. It limits the token selection to the smallest set whose cumulative probability exceeds **top-*p***. This truncates the tail of low-probability candidates, balancing output quality and diversity.

APPENDIX D. Pseudocode for entity redundancy estimation

Input:
entities = [e_1, e_2, …, e_n]//List of entities with embeddings
θ = *similarity_threshold*//e.g., 0.90
Step 1: Compute semantic similarity between all entity pairs
Initialize *similarity_graph* as empty undirected graph
For each pair (e_i, e_j) where $i < j$:
sim_score = SemanticSimilarity(e_i, e_j)//e.g., cosine similarity
If *sim_score* > θ:
Add edge between e_i and e_j in *similarity_graph*
Step 2: Identify connected components
components = FindConnectedComponents(*similarity_graph*)
Step 3: Count redundant entities
N_total_entities = len(*entities*)
N_entities_in_clusters = 0
For each *component* in *components*:
If size of *component* > 1:
N_entities_in_clusters + = size of component
Step 4: Compute redundancy score
RedundancyScore = *N_entities_in_clusters*/*N_total_entities*
Output:
RedundancyScore ∈ [0, 1]

APPENDIX E

The graph information entropy is calculated as:

$$H(G) = -\sum_{i=1}^{N} p_i \cdot log_2(p_i) \quad \text{(E -1)}$$

where $H(G)$ is the information entropy of graph G; N is the number of nodes in the KG; $p_i = d(i) \Big/ \sum_j d(i)$ refers to the normalized degree node i.

Data availability

Data will be made available on request.

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